

Question	Marking Guidance	Mark	Comments
06.1	1. (Rate of) transpiration/evaporation increases due to increased temperature OR (Rate of) transpiration/evaporation increases due to increased light intensity OR (Rate of) transpiration/evaporation increases due to decreased humidity OR (Rate of) transpiration/evaporation increases due to increased wind/air movement; 2. (So) increased kinetic energy (causing more water loss) OR (So) increased water potential gradient (so more water lost) OR (So) increased (water) diffusion gradient (so more water lost); 3. Stomata open (at sunrise/after 5 am) allowing gas exchange OR Stomata open (at sunrise/after 5 am) allowing carbon dioxide to enter; 4. (Some) stomata close at midday/after 11 am (reducing transpiration);	4 (4 x AO2)	Ignore reference to tide 1 and 2 Reject tide affecting transpiration/ water potential/ humidity 1 and 2 Correct link needed between factor affecting transpiration and the explanation 4 Accept at 11 am as the time when stomata close

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06.2	Correct answer for 2 marks, 6.6̇, 6.67 – 7 (%);; Accept for 1 mark, 0.05 (correct difference in transpiration rate) OR 6.6 (correct calculation, but incorrect rounding) OR 6.25/6.3 (correct calculation using incorrect denominator) OR 666/667 correct number sequence but decimal place in wrong place eg 66.7/0.0667 OR 0.75 as denominator	2 (2 x AO2)	

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06.3	Mark in groups, either 1 to 4 OR 5 to 8 1. Record mass/length before and after; 2. Place in sea water for (specified/equal) time; 3. Method to remove surface water; 4. Increase in mass/length shows water has been absorbed by osmosis OR Increase in mass/length shows cells have lower water potential; OR 5. Put tissue/cells on (microscope/glass) slide; 6. Add seawater (and leave) 7. Observe under (optical) microscope; 8. If cells become flaccid they do not have a lower water potential than seawater OR (If cells become) turgid cells show water is absorbed by osmosis OR (If cells become) turgid cells show cells have a lower water potential OR (If cells are) not flaccid/plasmolysis cells show water is not lost by osmosis OR (Determine) percentage plasmolysis;	4 (4 x AO3)	Accept 'weight' for 'mass'. Accept 'diameter' for 'length'. 2. Ignore period of time 2. Accept seawater in a dilution series Ignore blot dry before initial mass measurement Reject 'size' once then allow ECF. 3. Accept eg use tissue paper to dry OR blot dry 4. Accept root/mangrove for cells 8. Accept description of turgid (cells)

